

The rules behind choices

Lecture 01 • puzzles $\rightarrow$ logic $\rightarrow$ proof

Dr. Mudassir Shabbir

Associate Professor, CS

Bring curiosity. We supply the contradictions.



DO NOW • 60 SECONDS

Start before I start

ON YOUR SHEET

Write one thing that feels mathematical—but has no numbers.

A route, a rule, a ranking, a decision...



Today, in four moves

01

Introduction

02

Logistics

03

Three Problems

04

Logic

MEET YOUR INSTRUCTOR

Hello, I'm Mudassir.



DR. MUDASSIR SHABBIR

2014

Ph.D. + M.S.

Rutgers · New Jersey

NOW

Associate Professor

CS · LUMS · Lahore

FOCUS

Discrete math · GNNs · Networks

Discrete geometry

RECENTLY

Research Assistant Professor

Vanderbilt · Nashville

Dr. Mudassir

or

Professor

or

Prof.

or

Mudassir

Reaching Out



mudassir.shabbir@lums.edu.pk

SUBJECT: CS210 + Section + your subject

Always CC TF and HTA. Include section in the subject.

OFFICE HOURS

12:00-13:00 on TT

Slack from any updates.

ROOM

9-G47A

SSE

Rules of the game

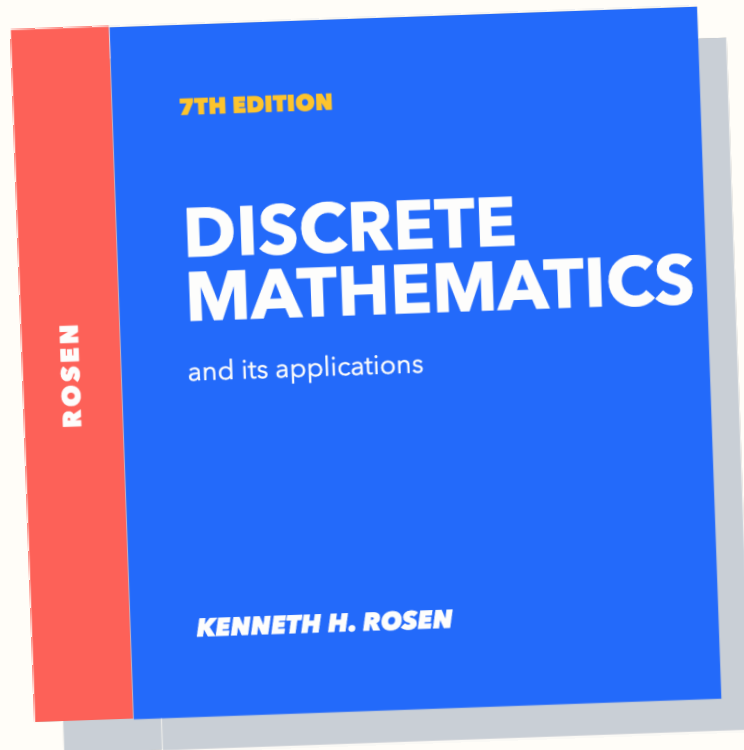
- 01** **ARRIVE IN THE PROBLEM** The mystery starts on time.
- 02** **PHONES FACE-DOWN** Keep your brain in the room.
- 03** **THINK OUT LOUD** Questions count as participation.
- 04** **BE WRONG HONESTLY** A clean mistake beats a lucky guess.
- 05** **ARGUE KINDLY** Attack the claim. Keep the human.



agreed? → prove it by participating

Whispering is just a podcast nobody subscribed to.

One useful book. Many excellent arguments.



McGraw-Hill

ISBN 978-0073383095

LIBRARY → a few copies

The book is heavy. The ideas travel light.

The SLACK is our course hub

SLACK • JOIN TODAY

tiny.cc/s2

the go-to place for everything, updates first

OFFICIAL LMS

lms.lums.edu.pk

materials • assignments • assessments

COURSE PAGE

amessbee.github.io/if26

syllabus • schedule • policies

COURSE RHYTHM

2 meetings each week

1 hour and 50 minutes each:

lecture + worksheet
in pairs
one answer sheet per group

ASSESSMENT

How points move



● WORKSHEETS

15%

● Long Quiz 1

25%

● Long Quiz 2

25%

● FINAL

35%

HW

are not graded

But excellent resource for learning and exam practice.

Worksheets: one class, two rhythms

1

**worksheet
every class**

WORK IN PAIRS

PHYSICAL NOTES

allowed

NO PHONES OR TABLETS

screens stay away

DISCUSS ONLY WITH YOUR PARTNER

Across-group discussion counts as plagiarism.

01

THE SPRINT

All questions at once · 15-20 minutes

02

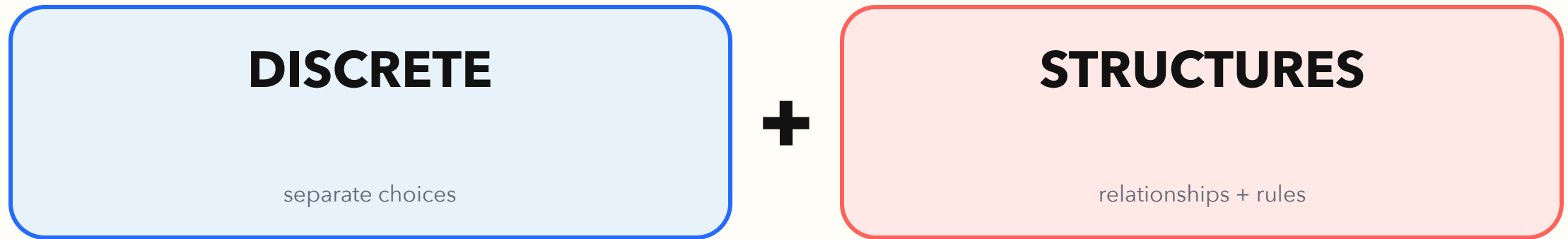
THE PULSE

One question at a time · short intervals throughout class

TODAY

WHAT IS THIS COURSE ABOUT?

Discrete + Structures



GRAPHS

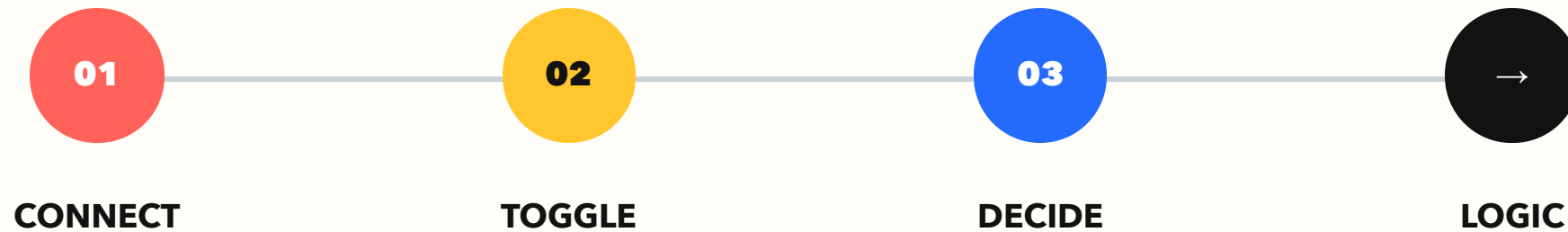
COUNTING

COMPUTATION

SETS

PROOFS

Three mysteries. One new language.



The sheet is your passport. Stamp it at every stop.

MYSTERY 01

Six people walk into a dinner...

Every pair has either met—or not.

SHEET M1

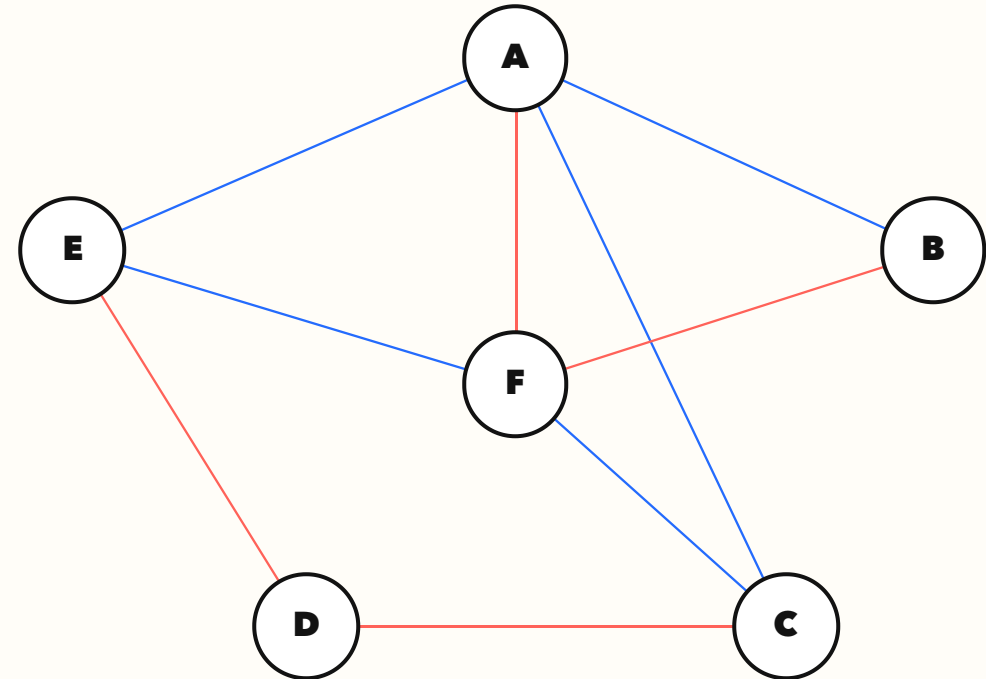


Must a matching trio exist?

Three mutual acquaintances or **three mutual strangers**

- YES – unavoidable
- NO – I can avoid it

Draw a counterexample—or explain why none can exist.

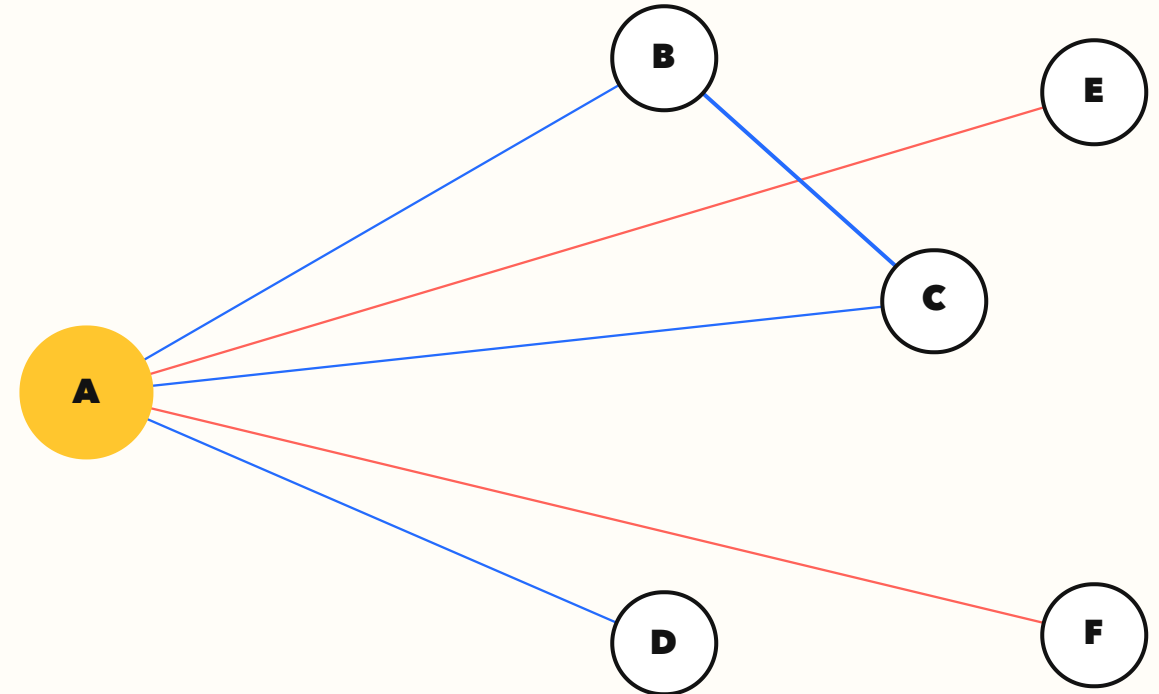


Pick one person. Three edges agree.

5 edges from A

At least 3 share a color.

If one inside edge matches: triangle.
If none match: the other color wins.

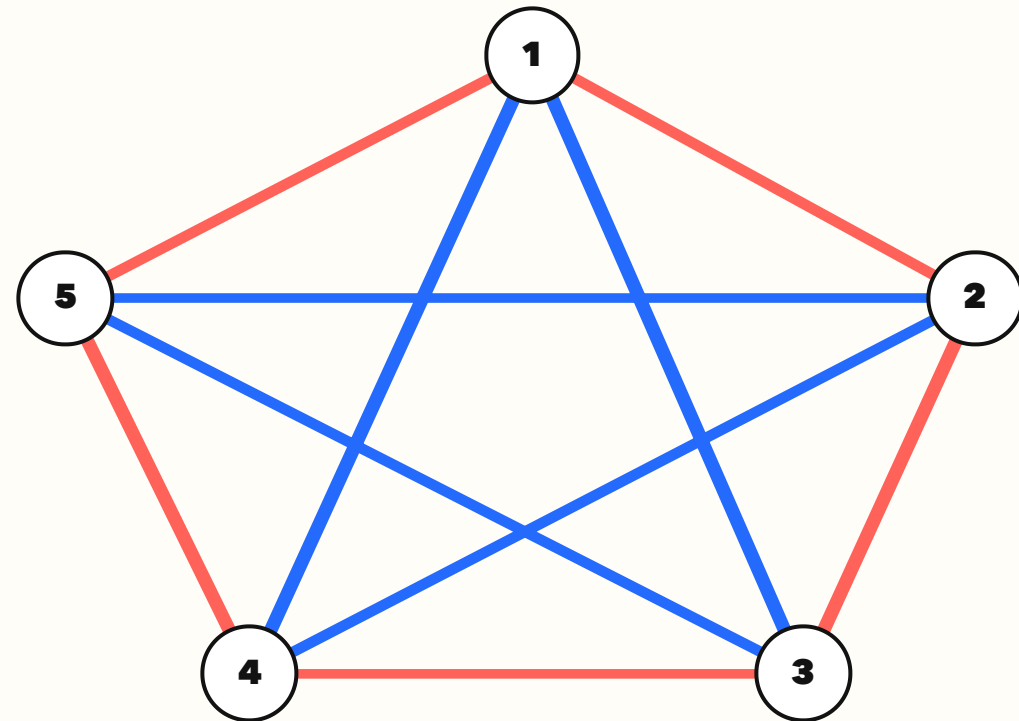


Six forces order. Five can still hide.

$$R(3,3) = 6$$

The first nontrivial Ramsey number.

Enough chaos eventually
contains a pattern.



WHY CARE?

A party puzzle became a network theorem



SOCIAL

**cliques +
communities**



COMPUTING

pattern detection



SCHEDULING

**conflict-free
groups**

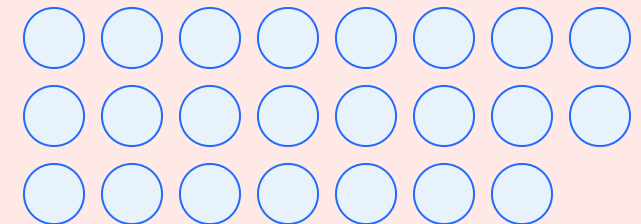
The picture changed the question. The proof changed what was possible.

MYSTERY 02

How crowded before birthdays collide?

More likely than not.

SHEET M2



VOTE FIRST

Would you bet on 23?

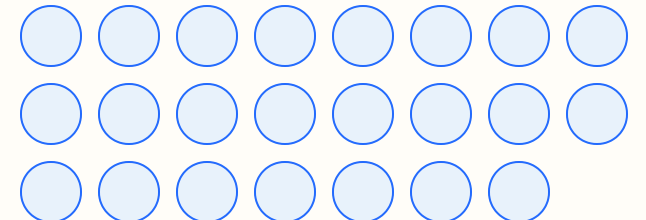
In a room of **23 people**...

$P(\text{shared birthday}) > 50\%$?

YES

NO

No calculators. Intuition must testify alone.



COUNT THE COMPLEMENT

Count the world with no collision

365/365

×

364/365

×

363/365

×

...

×

343/365

P(no match)

≈ 0.493

→

P(a match)

≈ 0.507

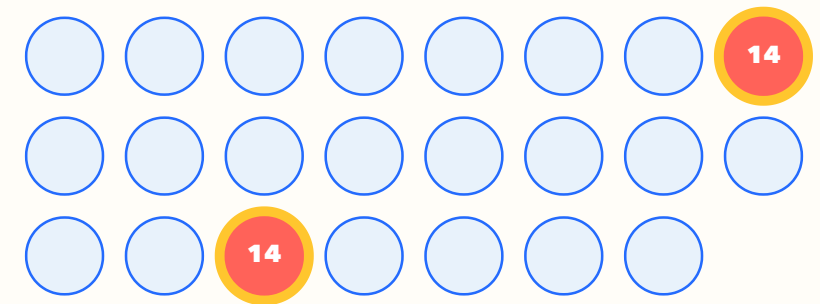
Sometimes the opposite event is easier to count.

THE SURPRISE

Twenty-three people create 253 pairs

$$23 \text{ choose } 2 = 253$$

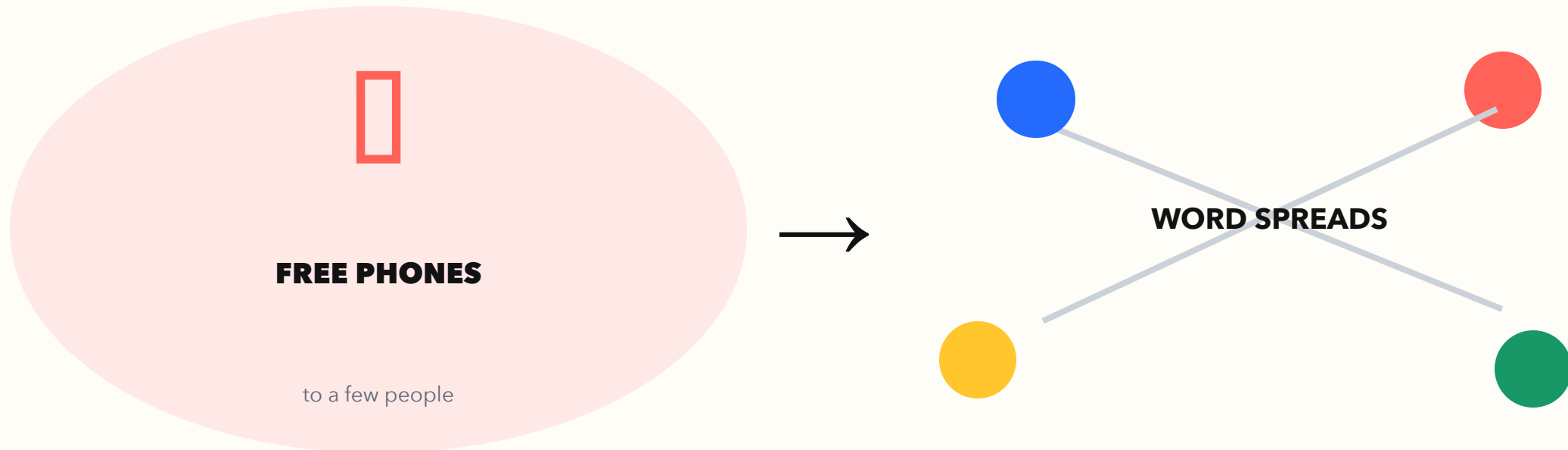
chances for a collision



Birthdays are independent. Party invitations are not.

MYSTERY 03

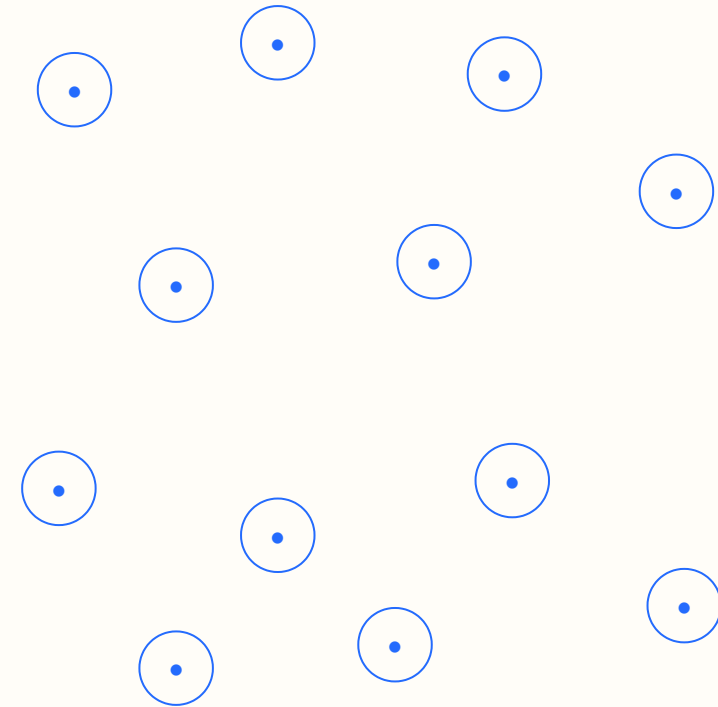
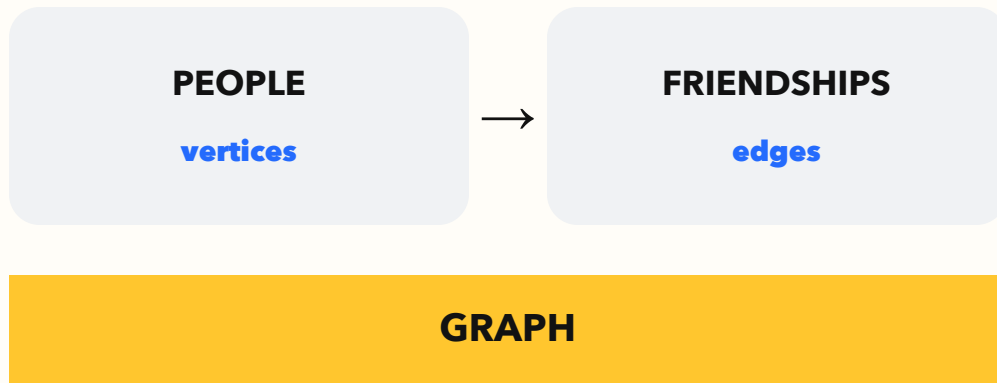
A phone company has a tiny budget.



SHEET M3

Minimum freebies. Whole community informed.

People become points. Friendships become lines.



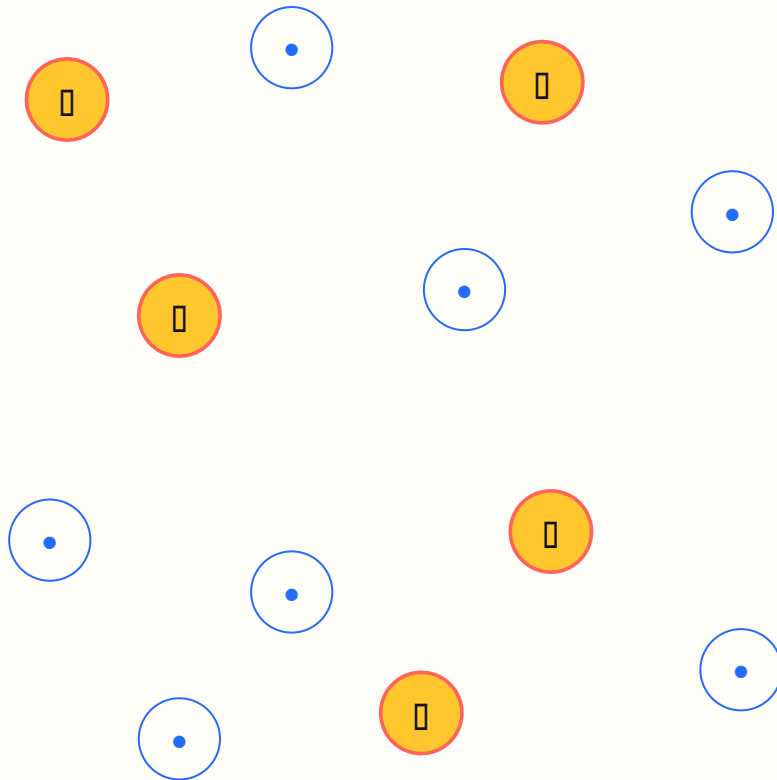
graphs?

counting?

sets?

proofs?

Five phones work. Is five clever?



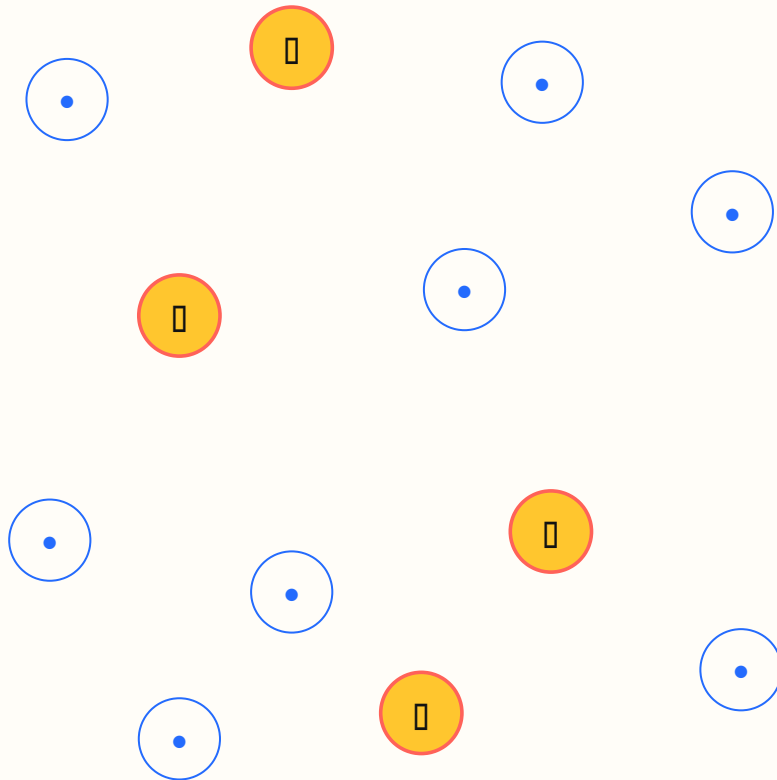
5

FREE PHONES

Every other person touches a recipient.

Can you do better?

Four phones work. Can three?



4

FREE PHONES

A smaller covering set.

On your sheet: what would prove 4 is best?

Different puzzles. Same mathematical habit.



CONNECT

people → graph

force a triangle



COUNT

matches → complement

253 pairs



COVER

network → minimum set

4 phones suffice

Next: make the arguments precise.

How do we make reasoning inspectable?

**FIRST STOP:
PROPOSITIONS**

Before we reason with a claim, we must know what kind of object it is.

01

PROPOSITIONS

Which sentences can be true or false?

02

CONNECTIVES

How do small claims combine?

03

ARGUMENTS

When does a conclusion really follow?

A PROPOSITION HAS ONE JOB

**Be true.
Or be false.**



No "kind of". No "on Tuesdays". Context first.